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Optimal power flow of a distribution system based on increasingly tight cutting planes added to a second order cone relaxation

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ABSTRACT

Convex relaxations of the optimal power flow (OPF) problem have received a lot of attention in the recent past. In this work, we focus on a second-order cone (SOC) relaxation applied to an OPF based on a branch flow model of a radial and balanced distribution system. We start by examining various sets of conditions ensuring the exactitude of such a relaxation, which is the main focus of the existing literature. In particular, we observe that these sets always include a requirement on the objective to be a minimization of a function increasing with the branch flow apparent powers. We consider this hypothesis to be at odds with what is to be expected of an active distribution system and demonstrate in specific case studies its counterproductive impact. We continue by introducing an objective function allowing distributed generations and storages (DGS) to take advantage of the benefits they bring to the power system as a whole. As this entails the possibility for the relaxation not to be exact, we describe and prove the theoretical convergence to optimality of an algorithm consisting in adding an increasingly tight linear cut to the SOC relaxation. In order to allow the attainment of a solution satisfying the network constraints in a finite number of steps, we continue by introducing a tailored termination criterion. Afterwards, we investigate the ability of our algorithm to obtain a satisfactory solution on several case studies, spanning various network sizes, number of nodes equipped with DGs and their level of penetration. We then conclude on the benefits brought about by this approach and reflect on its limits and the opportunities for further improvements.

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Introduction

With the aim to increase the sustainability of the electric power system, the share of renewable energies in the production mix is scheduled to increase in the future. For example, the European Union has set goals for its member states in order to attain a 20% share of renewable energy in its final energy consumption by 2020, and some countries have taken even more ambitious stances. This target will be partially met by integrating significant amounts of dispersed renewable energy generators (mainly photovoltaic (PV) and wind power) to the distribution grid. These developments will have a considerable impact on the design and operation of the electric system, both at the national and local level and so new tools will be needed to assist in the planning and operation of at least the distribution network. Indeed, as the current passive distribution network turns into an Active Distribution Network (ADN) (see [1] for a definition) with the introduction of partially and totally controllable generation and storage means, planning studies based solely on power flows for extreme load conditions will not be adapted anymore, as they would prevent the distribution system operators from taking advantage of the full benefits distributed generation could bring.

Considering the similarities between the current transmission network and the future ADN, it is a safe bet to assume that the optimal power flow, a framework first introduced in 1962 by [2] and now widely used for the planning and operation of the transmission network, will prove useful for this purpose. However, it has been remarked by several authors such as [3,4] that some characteristics inherent to the distribution system (in particular high R/X ratios and the radial nature of its topology) prevent us from applying traditional transmission system OPF algorithms, such as the Newton–Raphson method implemented in, for example, the Matpower package [5]. Moreover, simplifications commonly used in the planning of the transmission system, such as the linearization of the power flow constraints, are known to produce poor results with high R/X networks.

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In this context, the general objective pursued in this paper is to provide a methodology solving the single-stage OPF problem in a medium-voltage balanced radial distribution system to simulate DGS, with the purpose to integrate it in planning studies. This last conditions entails the need for a methodology that privileges speed of convergence and accuracy over precision, as it is expected to be run a large number of times to compare various planning options over long periods and with hour-long time steps, with significant uncertainties in input data. Additionally, its implementation should be versatile in terms of objective function and constraints to accommodate the uncertainties surrounding the future regulatory environment of the distribution system.

Of course, OPF in the distribution system have been envisioned for purpose other than the one described here, such as real time control [6] or optimal DG placement [7]. Nonetheless they all share an underlying common structure of Quadratically Constrained Quadratic Problem (QCQP) that we will describe below.

OPF as a QCQP

Traditionally, OPF based on bus injection model have been associated with meshed transmission systems and branch flow models have been used for radial distribution systems, see [8,9] for an up-to-date view on the subject. It has been recently proven that both can be cast as QCQP that are non-convex due to the presence of quadratic equality constraints, in the form of a rank constraint in the bus injection model [10] and power flow constraints in the branch flow model [11]. It is thus a NP-hard problem and cannot be solved in polynomial time while guaranteeing global optimality in general. A wide range of techniques have been employed including conjugate gradient, successive quadratic programming, branch-and-bound, Lagrange relaxation, interior point methods, simulated annealing, genetic algorithm and particle swarm optimization that represent various compromises between optimality and convergence speed and between versatility and tailoring for a specific problem (see [12] for a review of deterministic algorithms and [13] for non-deterministic ones). Considering our emphasis on convergence speed and versatility, and the parallel development of high performance SOC solvers and studies of quadratic convex relaxations of the OPF problem, we have chosen to focus our interest on them.

Convex relaxations

In the same manner that the equivalent QCQP of an OPF depends on which model is used, the corresponding convex relaxations will also depend on the model chosen.

1. Convex relaxation based on bus injection model are obtained by relaxing the rank constraint, as explained in [10]. We then obtain a semi-definite problem with a size proportional to the square of the size of the initial problem
2. Convex relaxation based on branch flow model. These relaxations are done in two steps. The first is a relaxation of bus angle constraints that is always exact for radial network [11] and the second relaxes quadratic equalities to convex inequalities to yield a SOC problem of a size equivalent to that of the original problem.

In the case of a radial network, [8] has shown that these relaxations are equivalent. Consequently, we choose the SOC option as the size of the resulting problem is smaller. This is supported by the findings in [8], where the authors have compared computational time to solve both relaxations on radial networks. While these were equivalent for small networks (9–39 nodes), the SOC relaxation was two orders of magnitude faster for the 300 bus test case.

Outline

For the remainder of this paper, we will focus on the SOC relaxation applied to a branch flow model of a balanced radial distribution system. We will start by stating a mathematical representation of the problem and present some of the conditions under which this relaxation may be exact, the focal point of most of the existing literature. We will then illustrate in which situations these conditions cannot be met on a simplified case study and introduce the cutting plane concept we will use in these instances, before proving its theoretical convergence to global optimality. We continue by introducing a termination criterion so that the algorithm is able to reach a satisfactory solution in a few iterations of a SOC solver, before analyzing its behavior on several case studies. We will conclude by reflecting on the applicability of the presented algorithm for our purpose and on the ways its performance and representativity of the problem can be enhanced.

The OPF model and its SOC relaxation

In the following, we will adopt the notations below:

Nomenclature

$j \in [1,J]$	the index of a node, where J is the node count and 1 is the root node
$f(j)$	the father node of node j, exists and is unique for all nodes except the root node
$c(j)$	the set of children nodes of node j
$P_{i,j}$	the active power flowing from node i to node j
$Q_{i,j}$	the reactive power flowing from node i to node j
$I_{i,j}$	the square of the current flowing from node i to node j
$r_{i,j}$	the resistance between node i and node j
$x_{i,j}$	the reactance between node i and node j
$z_{i,j}$	the impedance magnitude between node i and node j
$I_{i,j}^{max}$	the square of the maximal current that may flow from node i to node j
V_j	the voltage magnitude at node j
V_{OLTC}	the set of permissible voltages at the root node, dictated by the on-load tap changer characteristics
P_j^{st}	the active power injected by the storage inverter at node j
$P_j^{st,sp}$	the set point for the active power injected by the storage inverter at node j
Q_j^{st}	the reactive power injected by the storage inverter at node j
$S_j^{max,st}$	the maximal apparent power of the storage inverter at node j
P_j^{pv}	the active power injected by the PV inverter at node j
$P_j^{pv,sp}$	the set point for the active power injected by the PV inverter at node j
Q_j^{pv}	the reactive power injected by the PV inverter at node j
$S_j^{max,pv}$	the maximal apparent power of the PV inverter at node j
P_j^{load}	the active power load at node j
Q_j^{load}	the reactive power load j
$\mathbf{x}$	the vector of control and state variables, that thus includes active and reactive flows in the lines, voltage magnitudes at each nodes and active and reactive injections of storage and PV inverters
$\Delta(\mathbf{x})$	the convex quadratic objective function representing the interests of the DGS operators

Mathematical model

Drawing on the work of [14], we may formulate the OPF problem we wish to study with the following set of equations. Bus voltage angles are eliminated in accordance with [14].

The objective is:

$$\min \ \alpha \ \Delta(\mathbf{x}) + \beta \sum_{j\in[1,J]} r_{f(j),j} I_j \tag{1}$$

with $\alpha \geqslant 0$ and $\beta \geqslant 0$. It is thus a linear combination between a function representing the interests of the DGS operators and a function representing those of the distribution system operator (DSO) (e.g minimizing losses).

The branch flow model equations:

$$P_{f(j),j} - r_{f(j),j} I_{f(j),j} - \sum_{k\in c(j)} P_{j,k} - P_j^{load} + P_j^{st} + P_j^{pv} = 0 \tag{2a}$$

$$Q_{f(j),j} - x_{f(j),j} I_{f(j),j} - \sum_{k\in c(j)} Q_{j,k} - Q_j^{load} + Q_j^{st} + Q_j^{pv} = 0 \tag{2b}$$

$$U_{f(j)} - U_j - 2\left(r_{f(j),j} P_{f(j),j} + x_{f(j),j} Q_{f(j),j}\right) + z_{f(j),j}^2 I_{f(j),j} = 0 \tag{2c}$$

with:

$$I_{f(j),j} = \frac{{P_{f(j),j}}^2 + Q_{f(j),j}^2}{U_{f(j)}} \tag{3a}$$

$$U_j = V_j^2 \tag{3b}$$

We take into account the following constraints on the network:

$$I_{f(j),j} \leqslant I_{f(j),j}^{max} \tag{4a}$$

$$V_j^{min} \leqslant V_j \leqslant V_j^{max} \tag{4b}$$

$$V_1 \in V_{OLTC} \tag{4c}$$

Storage and PV power injections are constrained by their inverter apparent power:

$$P_j^{st2} + Q_j^{st2} \leqslant S_j^{max,st2} \tag{5a}$$

$$P_j^{pv2} + Q_j^{pv2} \leqslant S_j^{max,pv2} \tag{5b}$$

$$P_j^{pv} \in [0, P_j^{pv,sp}] \tag{5c}$$

$$P_j^{st} \in \begin{cases} [P_j^{st,sp}, S_j^{max,st}] & \text{if } P_j^{st,sp} < 0 \\ [-S_j^{max,st}, P_j^{st,sp}] & \text{if } P_j^{st,sp} > 0 \end{cases} \tag{5d}$$

In this representation, the non-convexity is concentrated in (3a) in the form of quadratic equalities. In the same manner as [14], we relax these to convex quadratic inequalities (6) to obtain the SOC relaxation.

$$I_{f(j),j} U_{f(j)} \geqslant P_{f(j),j}^2 + Q_{f(j),j}^2 \tag{6}$$

In the remainder of the paper, we will simplify the notation by replacing the index $f(j),j$ by j as it is unequivocal in radial networks.

Mathematical Conditions ensuring the exactness of the relaxation

Recently, the study of convex relaxations to the – distribution system – OPF has received considerable attention. Practically all this attention has been directed at establishing various sets of conditions on the objective function and the feasible injection region under which it is exact and we can cite [14–19] as prime examples of such a focus. In one of the two most encompassing approach for our purpose, [19] proves, for tree networks, that the Pareto-front of the feasible injection region is the same as that of its SOC relaxation. Consequently, if the objective function is convex and monotonically increasing in each active power injection and the initial OPF feasible, the relaxation will be exact. In parallel and using a different mathematical approach, it is proven in [18] that, if the objective function is strictly increasing in the active power injected at the substation and there can be no binding upper bound on voltages (and the mild condition C1 holds), the relaxation will be exact.

Examples of inexact relaxations in a simplified case study

In this section, we will test various instances of the objective function in simple case studies based on a 10 bus network and try to provide a practical understanding of the condition ensuring the exactness of the relaxation and explain why that may be antagonistic with the operation of the future ADN, which will justify the focus of this work on inexact relaxations. The topology and line characteristics of the 10 bus network are available in Table 1. One time step is modeled, with the active and reactive loads in Table 2. We study four different configurations in terms of distributed generation or storage that are described in Table 3.

For our present purpose, we consider the root node to have a fixed voltage magnitude of 1.03 p.u. and the lower and upper bounds on bus voltage magnitude to be 0.95 and 1.05. We also make the hypothesis

$$\Delta(\mathbf{x}) = \sum_{j\in[1,J]} |P_j^{pv,sp} - P_j^{pv}| + |P_j^{st,sp} - P_j^{st}| \tag{7}$$

meaning the DGS operator have a preferred set-point from which they want to minimize the absolute difference. Here, we would like to emphasize the fact that the absolute values in the objective function formulation, combined with the constraints on injections in Eq. (5), reduce to linear, and thus convex, functions of the injections.The three objective functions we study are then fully defined by the values taken by α and β:

1. $\alpha = 0$ and $\beta = 1$
2. $\alpha = 1$ and $\beta = 0$
3. $\alpha = 1$ and $\beta = 1$

The first objective function has the desirable property to be increasing over all the feasible injection region (the losses in the network are in fact the sum of all injections in the network, including the injection at the substation). Consequently, it can be used to determine the feasibility of an OPF problem, as it fulfills the requirement from [19]. However, in the presence of controllable active power injection in the form of storage devices or PV system that allow curtailment and that are large enough to create reverse power flows, minimizing losses can be contradictory with the objective of the storage or PV operator. Indeed, PV operators may be willing to allow the curtailment necessary to respect the network constraints if, for example, they are compensated by lower connexion costs. But curtailing beyond that level may be counterproductive, as the losses prevented will end up being only a fraction of the PV energy curtailed. The same reasoning can be

Table 1
Line parameters of the test network.

Line ID	$f(j)$	j	$r_{f(j),j}$	$x_{f(j),j}$
1	1	2	0.07882	0.04016
2	2	3	0.01072	0.03338
3	3	4	0.01829	0.06719
4	4	5	0.05393	0.04028
5	5	6	0.06239	0.07796
6	3	7	0.04172	0.00345
7	7	8	0.07030	0.04562
8	8	9	0.02258	0.00282
9	8	10	0.08972	0.05255

Table 2
Node parameters of the test network.

Node ID	P_j^{load}	Q_j^{load}
1	0	0
2	0.0168	3.10^{-4}
3	0.0197	4.10^{-4}
4	0.0186	5.10^{-4}
5	0.0162	5.10^{-4}
6	0.0150	4.10^{-4}
7	0.0277	4.10^{-4}
8	0.0223	3.10^{-4}
9	0.0297	5.10^{-4}
10	0.0262	3.10^{-4}

Table 3
Configurations of the test network that will be studied.

Parameters	Config. 1	Config. 2	Config. 3	Config. 4
Type	PV	PV	Storage	Storage
Node	3	3	4	4
Apparent Power (in p.u)	0.2	2	0.3	1
Set-point	0.2	2	−0.3	−1

applied to storage operators, considering the benefits they bring to the electric system as a whole over one cycle of operation.

This is the justification for the second objective function, for which only the point of view of the distributed generator operators is taken, with the aim to minimize the distance with their optimal set-points. The last objective function is a compromise between loss and set-point absolute distance minimization and so a compromise between the interests of the DSO and those of the distributed generator operators.

Results

For the remainder, we define the gap at the solution of the SOC in the following manner:

$$rGap(\mathbf{x}) = \sum_{j\in[1:N]} r_{f(j),j}\left(I_j U_{f(j)} - \left(P_j{}^2 + Q_j^2\right)\right) \tag{8}$$

We remark first that the line maximal intensity have been chosen not to be binding in any of the case studies, as focusing on the voltage constraints is the main issue for our present purpose. More precisely, as line intensity constraints are imposed on I_j, they will always be respected by the actual power flows resulting from the relaxation, its exactness notwithstanding.

To analyze the effect of the various objective functions, we will display five different types of result:

1. the gap, for each objective function and configuration in Table 4,
2. the power injected by either the storage or PV unit, at the optimal point of the relaxation, in Table 5,
3. the network active power losses, at the optimal point of the relaxation, in Table 6,
4. the voltages at each node at the optimal point of the relaxation (full line), called V_j^{SOC} hereafter, in Fig. 1,
5. the voltages at each node, obtained from a power flow with the active and reactive power injections fixed at their optimal value of the relaxation (dashed line), denominated V_j^{PF} in the remainder, in Fig. 1.

The SOC relaxation optimal solutions are calculated using the MOSEK optimization software [20] through its R [21] interface, with the relevant tolerances fixed at 10^{-5}. The fourth result could be obtained by solving a modified relaxation, minimizing the losses with the active and reactive power injections fixed in the same way and the first and second constraints in (5) removed. However it is faster to use a backward-forward sweep method based on Eqs. (2) and (3), provided it is, as the SOC solver, implemented in a compiled language.

Interpretation

We start by observing that, as expected for feasible OPFs, the relaxation is exact for the loss minimization objective in every case, and the losses at the optimal point are in fact substantially smaller than their initial value. We also note that the voltage constraints are far from being binding and that the active power injection have been modified heavily in some cases, thus hinting that PV and storage power has been curtailed more than is necessary to respect network constraints. Moreover, in configuration 2, PV has been curtailed by 1.83 p.u while the loss prevented (compared to their initial value) are only 0.46 p.u. The same situation is encountered for configuration 4, where storage has been curtailed by 1.19 p.u while the losses prevented are only 0.54 p.u.

Concerning the second objective function, we observe two distinct behaviors: when the active power injection set-point is positive (i.e for PV), the relaxation is inexact while it is exact when the injected active power set-point is negative (i.e charging of storage). This can be understood by noticing that, inside the feasible injection region, the objective function is increasing with the active power injection when the set-point is negative and decreasing otherwise. Consequently, the optimal solution will be in the pareto front of the feasible injection region and the relaxation exact only when all set-points for active power injection are negative.

The use of the third objective function brings a positive result in terms of exactness of the relaxation for configuration 1. To explain this, we can remark that, when the set-points are positive, and the injection cannot be higher than the set-point, minimizing the compromise function is in fact the same as minimizing the power injected at the substation because it is, algebraically, equal to the sum of active power injections and losses in the network:

$$P_{1,2} = \sum_{j\in[1\,J]} r_j I_j + \sum_{j\in[2,J]}\left(P_j^{load} - P_j^{st} - P_j^{pv}\right) \tag{9a}$$

$$= \sum_{j\in[1\,J]} r_j I_j + \sum_{j\in[2,J]}\left(|P_j^{pv,sp} - P_j^{pv}| + |P_j^{st,sp} - P_j^{st}|\right) + C \tag{9b}$$

Moreover, due to the low upper bound on injection in configuration 1, there can be no binding upper voltage constraints and condition

Table 4
Maximum gap (in p.u.) for various objective functions and configurations.

Objective	Config. 1	Config. 2	Config. 3	Config. 4
Loss minimization	0	0	0	0
Set-point absolute distance	0.01	0.12	0	0
Compromise	0	0.10	0	0

Table 5
Optimal active power injection for various objective functions and configurations.

Objective	Config. 1	Config. 2	Config. 3	Config. 4
Initial value	0.20	2.00	−0.30	−1.00
Loss minimization	0.17	0.17	0.19	0.19
Set-point absolute distance	0.20	2.00	−0.23	−0.54
Compromise	0.20	1.94	−0.23	−0.49

Table 6
Network Losses for various objective functions and configurations.

Objective	Config. 1	Config. 2	Config. 3	Config. 4
Initial Value	0.0028	0.4633	0.0527	0.5406
Loss minimization	0.0026	0.0026	0.0021	0.0021
Set-point absolute distance	NA	NA	0.0427	0.2315
Compromise	0.0028	NA	0.0427	0.1624

C1 from [18] holds, thus ensuring the exactness of the relaxation (see Theorem 1 from [18]). For configuration 2, the use of the compromise function only leads to a solution with a lower gap, without rendering it an exact relaxation because the upper bound on injection is high enough to entail binding upper voltage constraints. For configuration 3 the result is left unchanged, because, due to the absence of binding network constraints, the result was already a compromise. For configuration 4, the result is changed compared to the preceding objective function as the storage is curtailed by a further 0.05 p.u, allowing a decrease in losses of 0.07 p.u which is still an improvement in terms of overall energy efficiency.

We have shown that there exist situations where the interests of the DSO – minimizing losses and respecting network constraints – and the interests of the operator of the distributed generators and storages can be contradictory. In these situations, satisfying the interests of the DG operators (e.g minimizing the distance with the set points) may lead to inexact SOC relaxations. The use of the compromise objective function presented the obvious advantage of balancing the objective of the DSO with that of the DG and storage owners. Additionally, it allowed for an exact relaxation when set-points are positive and there can be no upper voltage binding constraint. Extracting a physically meaningful power flow respecting network constraints when power injection can lead to over-voltages will be the focus of the next section.

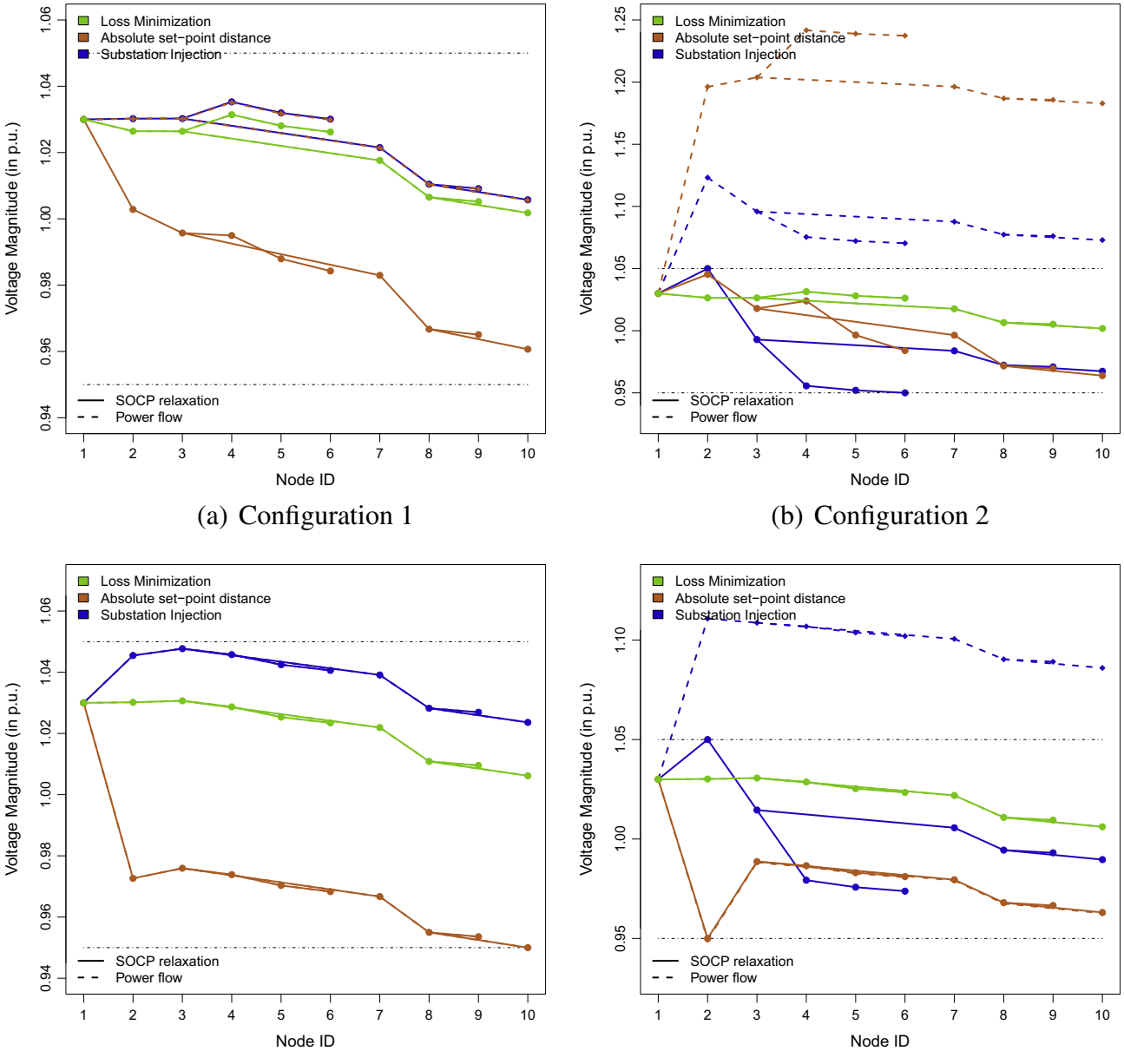


Fig. 1. Voltages obtained by SOCP relaxation and power flow for various objective functions.

Extracting a physically meaningful solution

Insofar as we tackle the case of inexact relaxations, our work is in the vein of the recent efforts by [22–25]. However, the comparison ends here as the four articles above address the more general problem of meshed networks, at the expense of the use of a more computationally demanding SDP relaxation. Moreover, our methodology relies on a cutting plane concept, while [22,23] propose a branch-and-bound approach, [24] develops a heuristic to extract rank-1 solutions from low-rank solutions and [25] uses an approximation of a rank-minimization algorithm.

In this section, we start by defining the methodology we implement and proving its convergence. We then illustrate it with detailed results on a simplified case study. By combining observations on these results with a qualitative assessment of the accuracy requirements of the expected OPF application, we subsequently propose a stopping criteria allowing the attainment of a satisfactory solution.

Principle of the cutting planes

The principle behind the use of cutting planes can be dated back to Gomory who first introduced them in 1958 to solve mixed integer linear programming [26]. The idea is to solve a relaxation of the original problem (for Gomory, ignoring the integer requirement, here, the SOC relaxation), then, if the relaxation is not exact, adding a linear inequality to the relaxation that will eliminate the first solution obtained from the feasible set while keeping the optimal solution of the non-relaxed problem in the feasible set. Then this method is applied iteratively and new cuts are added at each step, until a termination criterion is met. Of course, the complexity of such a methodology lies in finding a way to calculate the cutting planes that is computationally tractable and fulfills the requirements mentioned above and we will now describe the approach we put forward.

Definition of the cuts

To improve the readability of the following section, we define S as the set of feasible solutions to the original problem, $\tilde{S}_k$ as the set of feasible solutions of its SOC relaxation minus the points eliminated by the cut imposed at step k, starting at step 0 with no cut. Moreover:

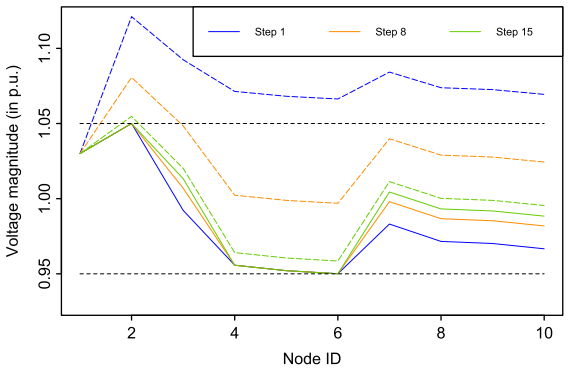


Fig. 3. Illustration of the algorithm on configuration 2.

$$rI(\mathbf{x}) = \sum_{j\in[1,J]} r_{f(j),j} I_j \tag{10a}$$

$$rL(\mathbf{x}) = \sum_{j\in[1,J]} r_{f(j),j} \frac{P_j^2 + Q_j^2}{U_{f(j)}} \tag{10b}$$

$$\dot{\mathbf{x}} = \underset{\mathbf{x}\in S}{\operatorname{argmin}}\ \alpha\, \Delta(\mathbf{x}) + \beta\, rI(\mathbf{x}) \tag{10c}$$

$$\tilde{\mathbf{x}}_k = \underset{\mathbf{x}\in \tilde{S}_k}{\operatorname{argmin}}\ \alpha\, \Delta(\mathbf{x}) + \beta\, rI(\mathbf{x}) \tag{10d}$$

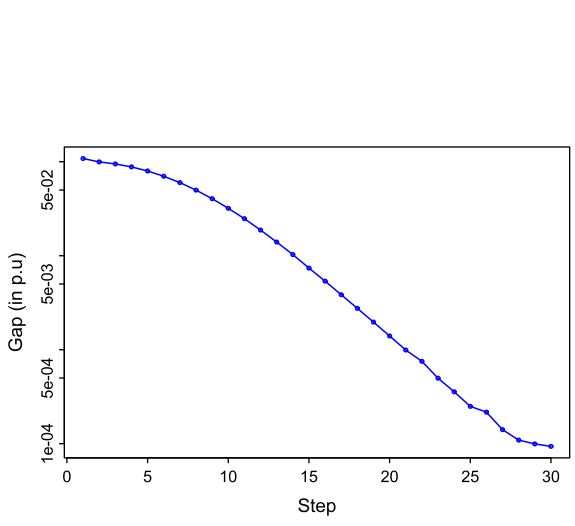
To focus ourselves on interesting situations, we make two assumptions in the remainder:

1. S is not empty,
2. the solution obtained at step k is not exact,
3. $\alpha > 0$ (otherwise the relaxation would be exact),
4. $\beta > 0$.

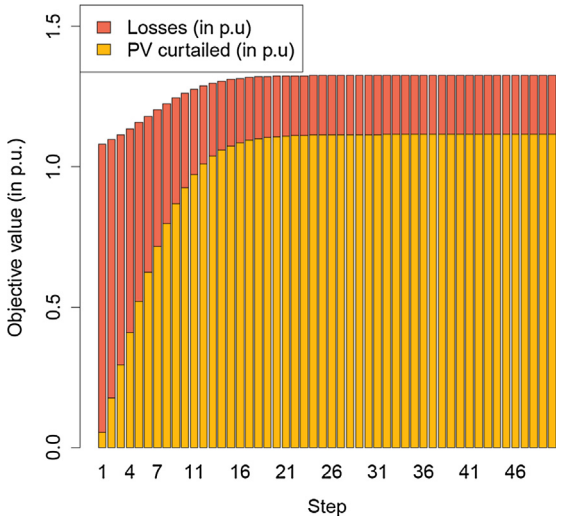
We then define the cut at step k in the following manner:

$$rI(\mathbf{x}) \leqslant rL(\tilde{\mathbf{x}}_{k-1}) \tag{11}$$

It is straightforward to see that the current optimal point is eliminated from the feasible (the second assumption ensures that



(a) Gap reduction



(b) Objective value evolution

Fig. 2. Illustration of the algorithm on configuration 2.

Table 7
Trade-off between computation time and optimality, with a stopping criterion allowing the achievement of a solution respecting network constraints.

ϵ_{Cons}	5.10^{-4}	1.10^{-3}	5.10^{-3}	1.10^{-2}	15.10^{-3}
Number of Steps	24	20	14	12	10
objective value	1.33	1.34	1.37	1.41	1.47

$rL(\tilde{\mathbf{x}}_{k-1})$ is strictly inferior to $rI(\tilde{\mathbf{x}}_{k-1})$) and thus the next subsection will consist in proving that this cut keeps $\dot{\mathbf{x}}$ in $\tilde{S}_k$.

Proof of the validity of the cuts

What we know from the existing literature is that minimizing the losses over $\tilde{S}_{k-1}$ will lead to a solution in S, provided that the cut at step $k-1$ is valid. Moreover, we have observed that the objective of set-point distance minimization can be contradictory to the objective of loss minimization and thus lead to an inexact relaxation. Intuitively, what happens in such situations is that we have not reduced the losses "enough" to allow for a feasible solution to be obtained, which is the informal justification for adding cuts of the form (11). We will now formally prove that the cut defined in (11) is valid for all step k by induction. The base case for $k=0$ is plain, as $S \subset \tilde{S}_0$. The fact that the validity of the cut at step $k-1$ entails the validity at step k is proven in three step:

$$rI(\dot{\mathbf{x}}) < rI(\tilde{\mathbf{x}}_{k-1}) \tag{12a}$$
$$\Delta(\dot{\mathbf{x}}) \geqslant \Delta(\tilde{\mathbf{x}}_{k-1}) \tag{12b}$$
$$rI(\dot{\mathbf{x}}) < rL(\tilde{\mathbf{x}}_{k-1}) \tag{12c}$$

Step 1. We prove (12a) by contradiction. Let us assume that $rI(\dot{\mathbf{x}}) \geqslant rI(\tilde{\mathbf{x}}_{k-1})$. Due to theorem 8 in [19] (supposing that Assumption 2 is met) and the validity of the cut at step $k-1$, we know the following:

$$\exists! \hat{\mathbf{x}} \in S : \hat{\mathbf{x}} = \underset{\substack{\mathbf{x}\in\tilde{S}_{k-1} \\ rI(\mathbf{x})\geqslant rI(\tilde{\mathbf{x}}_{k-1})}}{\operatorname{argmin}} rI(\mathbf{x}) \tag{13}$$

We also have:

$$\tilde{\mathbf{x}}_{k-1} = \underset{\substack{\mathbf{x}\in\tilde{S}_{k-1} \\ rI(\mathbf{x})\geqslant rI(\tilde{\mathbf{x}}_{k-1})}}{\operatorname{argmin}} rI(\mathbf{x}) \tag{14}$$

Due to the uniqueness in (13), we obtain $rGap(\hat{\mathbf{x}}) = rGap(\tilde{\mathbf{x}}_{k-1})$, which is in contradiction to the second assumption and the fact that $\hat{\mathbf{x}}$ belongs to S.

Step 2. If the cut at step $k-1$ is valid, $\dot{\mathbf{x}}$ belongs to $\tilde{S}_{k-1}$. $\dot{\mathbf{x}}$ cannot have an objective value that is lower than that of $\tilde{\mathbf{x}}_{k-1}$, as this would contradict the definition of $\tilde{\mathbf{x}}_{k-1}$ and the assumption of the inexactness of the relaxation. Consequently, we have:

$$\alpha(\Delta(\dot{\mathbf{x}}) - \Delta(\tilde{\mathbf{x}}_{k-1})) \geqslant \beta(rI(\tilde{\mathbf{x}}_{k-1}) - rI(\dot{\mathbf{x}})) \tag{15}$$

Combining this with (12a), we obtain (12b)

Step 3. $\tilde{\mathbf{x}}_{k-1}$ is a pareto efficient solution for the minimization of the vector valued objective function:

$$x \rightarrow (\Delta(\mathbf{x}), rL(\mathbf{x}), rGap(\mathbf{x})) \tag{16}$$
$$x \in \tilde{S}_{k-1} \tag{17}$$

because it is the solution of a minimization of a linear combination of the element of the above function. By definition of the pareto efficient solutions, we have:

$$rGap(\tilde{\mathbf{x}}_{k-1}) = \underset{\substack{\mathbf{x}\in\tilde{S}_{k-1} \\ rL(\mathbf{x})\leqslant rL(\tilde{\mathbf{x}}_{k-1}) \\ \Delta(\mathbf{x})\leqslant\Delta(\tilde{\mathbf{x}}_{k-1})}}{\operatorname{argmin}} rGap(\mathbf{x}) \tag{18}$$

Due to the fact that $rGap(\mathbf{x})$ cannot be negative and $\dot{\mathbf{x}}$ belongs to $\tilde{S}_{k-1}$, we have:

$$rGap(\dot{\mathbf{x}}) = 0 = \underset{\substack{\mathbf{x}\in\tilde{S}_{k-1} \\ rL(\mathbf{x})\leqslant rL(\dot{\mathbf{x}}) \\ \Delta(\mathbf{x})\leqslant\Delta(\dot{\mathbf{x}})}}{\operatorname{argmin}} rGap(\mathbf{x}) \tag{19}$$

From the second assumption, we know that $rGap(\dot{\mathbf{x}}) < rGap(\tilde{\mathbf{x}}_{k-1})$. We combine this with Eq. (12b) to obtain:

$$rL(\dot{\mathbf{x}}) < rL(\tilde{\mathbf{x}}_{k-1}) \tag{20}$$

As, by definition of $\dot{\mathbf{x}}$, $rGap(\dot{\mathbf{x}}) = 0$, we have demonstrated (12c) and the validity of the cuts at each step k.

Illustration of the convergence on configuration 2

We start by applying our methodology to the case study defined above for which the relaxation is inexact. The results provided in Fig. 2 for the first 30 steps are $rGap(\tilde{\mathbf{x}}_k)$, $rI(\tilde{\mathbf{x}}_k)$ and $\Delta(\tilde{\mathbf{x}}_k)$. $rGap(\tilde{\mathbf{x}}_k)$ is plotted in Fig. 2(a) with a logarithmic scale. $rI(\tilde{\mathbf{x}}_k)$ and $\Delta(\tilde{\mathbf{x}}_k)$ are represented as a stacked bar graph in Fig. 2(b), with the total height of the bars being the objective value at the considered step. First, we can observe that $rGap(\tilde{\mathbf{x}}_k)$ decreases consistently during the process while the objective value of the relaxation converges asymptotically to the optimal value of the original problem. Moreover, the trade-off between losses and set-point deviations evolves as expected, with high losses and low set-point deviations for the first steps and conversely for the last ones.

In a manner comparable to Section "Results", we further illustrate the convergence properties of our algorithm by drawing V_j^{SOC} and V_j^{PF} for steps 1, 8 and 15 in Fig. 3. In coherence with the continuous decrease of $rGap(\tilde{\mathbf{x}}_k)$, we witness that V_j^{SOC} and V_j^{PF} get closer to each other as the algorithm approaches the optimal solution. We must now emphasize the fact that, while the algorithm converges to the optimal solution of the original problem, each solution obtained will necessarily lead to over-voltages, as illustrated in Fig. 3. Consequently, obtaining a solution that satisfies the network constraints in a finite number of steps will require a compromise between closeness to the optimal solution and computational cost that will be the subject of the next section.

Stopping criterion

To overcome the fact that the injection set points obtained will lead to over-voltages, we propose to decrease the upper voltage bound by a given ϵ, and to stop the process when the original voltage bound is respected. With this procedure, network constraints are guaranteed to be respected in a finite number of steps, at the expense of an over-valuation of the objective function. The choice of ϵ will thus be a trade-off between execution time and optimality, as illustrated in Table 7, with high values of ϵ leading to a fast solution with a high objective value and conversely for small ϵ.

Table 8
Characteristics of the case studies.

Case ID	1	2	3	4	5	6	7	8	9	10	11	12
Network size	10				69				189			
Total load (in p.u.)	0.2				13				13.8			
Buses Equipped	2		5		14		35		38		95	
Total PV (in p.u)	1.2	2.5	2.4	2.9	64	86	85	171	133	222	388	776

Table 9
PV curtailment for the SOCP relaxation without cuts, and for two values of the stopping criteria.

Case ID	1	2	3	4	5	6	7	8	9	10	11	12
SOCP relaxation (%)	5.2	34	2.0	5.6	24	22	21	26	20	25	29	39
$\epsilon = 5.10^{-3}$ (%)	20	65	28	32	27	32	29	45	71	35	38	53
$\epsilon = 1.10^{-2}$ (%)	27	71	31	34	29	36	33	51	74	37	40	57

Table 10
$\max V_j^{PF}$ for the SOCP relaxation without cuts, and for two values of the stopping criteria.

Case ID	1	2	3	4	5	6	7	8	9	10	11	12
SOCP relaxation	1.088	1.192	1.210	1.233	1.071	1.143	1.106	1.370	1.070	1.110	1.211	1.453
$\epsilon = 5.10^{-3}$	1.050	1.050	1.050	1.050	1.050	1.050	1.050	1.050	1.048	1.049	1.050	1.050
$\epsilon = 1.10^{-2}$	1.049	1.049	1.050	1.050	1.050	1.050	1.050	1.050	1.047	1.049	1.050	1.050

Table 11
Objective value for the SOCP relaxation without cuts, and for two values of the stopping criteria.

Case ID	1	2	3	4	5	6	7	8	9	10	11	12
SOCP relaxation	0.54	1.71	1.48	1.95	29.4	44.0	39.0	98.5	45.7	99.3	213.7	533.6
$\epsilon = 5.10^{-3}$	0.60	1.85	1.54	2.00	30.4	45.2	40.3	101.4	49.6	104.4	220.7	547.1
$\epsilon = 1.10^{-2}$	0.65	1.90	1.58	2.03	30.9	45.7	40.8	101.5	53.7	109.3	225.8	550.7

Table 12
Mean duration of individual SOCP steps and number of SOCP steps before termination, for two values of the stopping criterion.

Case ID	1	2	3	4	5	6	7	8	9	10	11	12
Mean step duration (in ms)	9	10	11	9	26	27	26	26	52	53	57	54
$\epsilon = 5.10^{-3}$	9	13	108	118	35	61	49	85	3	10	18	51
$\epsilon = 1.10^{-2}$	7	9	25	74	16	17	18	37	3	8	4	19

Case study

Definition of the configurations studied

First, we must make clear that we agree wholeheartedly with the claim in [27] that practical systems operating at normal conditions will exhibit exact relaxations for the associated OPF models and so we do not intend to study them. In other words, the case studies we present here have an overt bias towards dealing with very large active power injections at times of low loading in future hypothetical distribution systems with very high levels of distributed generation penetration. The objective of such a focus in this section is not to draw conclusions on the necessity of PV curtailing or storage development in general, but rather to assess the ability of our algorithm to attain satisfactory solutions when some characteristics of the input that we deem relevant are modified. These characteristics are: the number of nodes in the network, the number of nodes equipped with PV and the ratings of the PV inverters. The network data for the considered case studies are based on the 10-bus network described above, the 69-bus network in [28] and the 189-bus network in [29]. For each network, we study four configurations of PV injection (two number of nodes equipped and two combinations of PV inverter ratings). The buses equipped are selected in decreasing order of apparent load, again in the spirit of studying the worst conceivable situations. Moreover, the active power injection set-points are assumed to be at the maximal value allowed by the inverter rating, the voltage bounds are ±5% around the nominal value and the voltage at the root node is fixed at +3% of the nominal value. Table 8 details these twelve configurations.

Analysis of the results

The results are presented in Tables 11, 10 and 9, where are displayed, respectively, the objective value, $\max V_j^{PF}$ and the percentage of PV curtailed for the SOC relaxation without cuts and at the termination step for two settings of ϵ. Table 12 displays the mean duration of a single SOC optimization for each case, along with the number of steps necessary to fulfill the termination criterion.

We can first remark that, at constant network size, the number of node equipped has no influence on the duration of an individual step – variation on the order of the millisecond should not be over-interpreted as these are close to the measurement precision. We also observe that, across the various network sizes, the variation in individual step duration is fairly limited, as multiplying the network size by, respectively, 7 and 19 leads to an increase in duration by a factor of, respectively, around 2.5 and 5.5, which is very encouraging in terms of scalability of our approach. To be clear, this is not a feature of our algorithm *per se* but rather a consequence of being able to draw on the performance of existing SOC solvers. Concerning the ability of our algorithm to fulfill its purpose, we may start by observing that a satisfactory solution in terms of network constraints has been obtained in a finite number of steps for each of the case study and termination settings, as the results in Table 10 exhibit no remaining over-voltages. Moreover, the results in Tables 9 and 11 are unsurprising, with more stringent termination criterion leading to better solution in terms of objective function or PV curtailing and increased level of penetration – all things else equal – leading to higher curtailing. However, when trying to draw conclusion on a potential link between the input data and the number of steps, the situation is more muddled.

While, as could be expected, more stringent criteria, higher level of penetration or number of nodes equipped lead, for a given network, to higher number of steps needed, the variation in the amount of steps between the various network does not seem to be linked to the number of nodes in the network. This may be discomfiting at first but might be explained by the fact that there are many parameters in the network definition that can influence the efficiency of our algorithm that we do not isolate in these case study, such as the R/X ratio of the lines, the dispersion of the value of the loads and PV generator injections or the position of such generators relative to the root node. Consequently, additional investigations of this particular issue are in order.

Conclusion and perspective

After reviewing the existing literature that focuses on proving the existence of conditions ensuring the exactness of the SOC relaxation of the distribution system optimal power flow, we have illustrated how these could be incompatible with the efficient operation of future distribution systems with high penetration of distributed resources. We have then introduced an algorithm consisting in iteratively adding linear cuts to this SOC relaxation, before proving its theoretical convergence to the optimal solution. We continued by illustrating its practical implementation on various case studies while proposing a termination criterion allowing the achievement of a solution satisfying the network constraints in a finite number of steps. In the worst situation (biggest network, highest number of nodes equipped, highest penetration and best precision), a satisfactory solution was obtained in around 2.5 s. Considering the fact that, in most cases, the number of hours in a year for which DGS injections will be high enough to force inexact relaxations will be low compared to the total number of steps, we can assume that the calculation of the objective value for a given set of planning options could be done in a few minutes, which is a promising result considering our initial objective. In the short term, our future work on the subject will focus on two areas. First, extending the range of our case studies in order to better understand the link between the characteristics of a given problem and the number of step needed to satisfy a given termination criterion. Secondly, developing a tailored scaling method that would allow us to improve the accuracy settings of the underlying SOC solver. In the long term, possible developments will consist in tackling both the issues of three-phase unbalanced networks and multistage optimal power flow.

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